2025 VCE Specialist Mathematics Year 12 Trial Examination 1

Kilbaha Education

Quality educational content

Kilbaha Education (Est. 1978) (ABN 47 065 111 373) **PO Box 3229** **Cotham Vic 3101** **Australia**	**PayID: 47065111373** **Email: kilbaha@gmail.com** **Tel: +61 413 425 374** **Web: https://kilbaha.com.au**

This publication from Kilbaha Education is digital and is supplied to the purchasing school in both WORD and PDF formats with a school site licence to reproduce for students in both print and electronic formats.

Kilbaha Education (Est. 1978) (ABN 47 065 111 373)
PO Box 3229
Cotham Vic 3101
Australia
PayID: 47065111373
Email: kilbaha@gmail.com
Tel: +61 413 425 374
Web: https://kilbaha.com.au

IMPORTANT COPYRIGHT NOTICE FOR KILBAHA PUBLICATIONS
(1) The material is copyright. Subject to statutory exception and to the provisions of the relevant collective licensing agreements, no reproduction of any part may take place without the written permission of Kilbaha Pty Ltd.
(2) The contents of these works are copyrighted. Unauthorised copying of any part of these works is illegal and detrimental to the interests of the author(s).
(3) For authorised copying within Australia please check that your institution has a licence from https://www.copyright.com.au This permits the copying of small parts of the material, in limited quantities, within the conditions set out in the licence.
(4) All pages of Kilbaha files must be counted in Copyright Agency Limited (CAL) surveys.
(5) Kilbaha files must not be uploaded to the Internet.
(6) Kilbaha files must not be used to train Artificial Intelligence (AI) Chatbots.
(7) Kilbaha files may be placed on a password protected school Intranet.

Kilbaha educational content has no official status and is not endorsed by any State or Federal Government Education Authority.

While every care has been taken, no guarantee is given that the content is free from error. Please contact us if you believe you have found an error.

Kilbaha Education (Est. 1978) (ABN 47 065 111 373) **PO Box 3229** **Cotham Vic 3101** **Australia**	**PayID: 47065111373** **Email: kilbaha@gmail.com** **Tel: +61 413 425 374** **Web: https://kilbaha.com.au**

Write your **student number** in the boxes above. **Letter**

Specialist Mathematics Examination 1

Question and Answer Book VCE Examination

- Reading time is **15 minutes**:
- Writing time is **1 hour**:

Materials supplied

- Question and Answer Book of 22 pages.
- Formula Sheet

Instructions

Students are **not** permitted to bring any technology (calculators or software), or any notes of any kind into the examination room.

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

Contents pages

10 questions (40 marks) _____ 4-15

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Instructions

- Answer **all** questions in the spaces provided.
- Write your responses in English
- In all questions where a numerical answer is required, an **exact** value must be given unless otherwise specified.
- In questions where more than one mark is available, appropriate working **must** be shown.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
- Take the acceleration due to gravity to have a magnitude $g \text{ ms}^{-2}$, where $g = 9.8$

Question 1 (3 marks)

Solve the equation $\cos(2x) = |\cos(x)|$ where $x \in R$.

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Question 2 (2 marks)

Prove that $\log_2(3)$ is irrational.

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Question 3 (4 marks)

Let $f : R \setminus \{1\} \to R,\ f(x) = \dfrac{x^2 - 4}{x - 1}$

a. Determine the equations of all the straight line asymptotes on the graph of $y = f(x)$.

1 mark

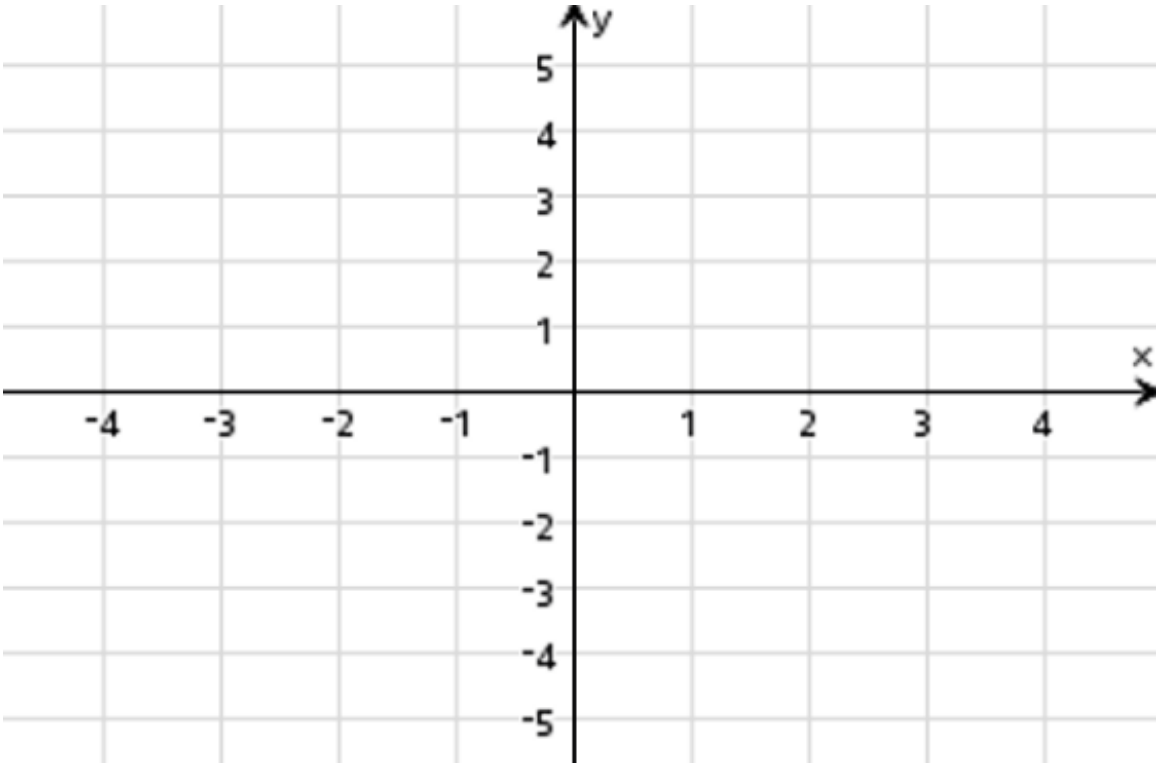
b. Determine whether or not the graph of $y = f(x)$ has any turning points or points of inflection.

1 mark

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

c. Sketch the graph of $y = f(x)$ on the axes below. Label the asymptotes with their equations and the axial intercepts with their coordinates.

2 marks



© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Question 4 (6 marks)

A particle moves in a straight line. The displacement from a fixed origin is x metres, at a time t seconds, its velocity $v\,\mathrm{m\,s^{-1}}$ is given by $v=\dfrac{1}{x\left(4+t^2\right)}$ and when $t=2$, $x=-1$.

a. Find the acceleration of the particle, in $\mathrm{m\,s^{-2}}$ when $t=2$ and $x=-1$.

3 marks

b. Express x in terms of t.

3 marks

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Question 5 (4 marks)

The curve defined by the parametric equations $x = 2t^3$, $y = 3t^2$ for $0 \le t \le 2$ is rotated about the x-axis to form a surface of revolution. The surface area can be represented by a definite integral $k\pi\int_a^b (cu - a)\sqrt{u}\,du$. Find the values of k, a, b and c.

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Question 6 (3 marks)

The weights of packages of T-bone steaks are normally distributed with a mean of 350 grams and standard deviation of 4 grams. The T-bone steak actually contains a bone. The weights of the bone are normally distributed with a mean of 60 grams and standard deviation of 2.5 grams. The steak also contains some fat. The fatty bits are normally distributed with a mean of 20 grams and standard deviation of 1.5 grams. The edible part of the T-bone is the remaining weight without the bone and without the fat.

a. Determine the mean and standard deviation of the edible part of **two** packages of T-bone steaks.

2 marks

b. Given that $\Pr(-2 \le Z \le 2) = 0.95$, where Z has the standard normal distribution, determine the probability that the edible part of two packages of T-bone steaks is more than 526 grams, give your answer correct to three decimal places.

1 mark

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Question 7 (4 marks)

Prove by induction that $3^{2n+1}+2^{n+2}$ is divisible by 7 for $n \in N$.

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Question 8 (6 marks)

Let $w = 1 + yi$ where $w \in C$, $y \in R^+$ and $\text{Im}\left(w^5\right) = 0$.

a. Express w in polar form and hence determine all possible values of y.

3 marks

b. Given that $(1+x)^5 = 1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5$ express w^5 in rectangular form and hence find $\tan\left(\frac{\pi}{5}\right)$ giving your answer in the form $\sqrt{a + b\sqrt{a}}$, where $a, b \in Z$.

3 marks

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Question 9 (5 marks)

a. Evaluate $\int_0^1 \frac{x}{(x+1)^2}\,dx$.

1 mark

b. Let $I_n = \int_0^1 \frac{x^n}{(x+1)^2}\,dx$. Show that $I_n = \frac{1}{2(n-1)} - \frac{n}{n-1} I_{n-1}$ for $n \geq 2$.

3 marks

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

c. Hence find $\int_0^1 \frac{x^3}{(x+1)^2}\,dx$

1 mark

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Question 10 (3 marks)

a. Write down the Cartesian equation of the plane which crosses the x-axis at $x = 4$, the y-axis at $y = -3$ and the z-axis at $z = 6$.

1 mark

b. Find the values of x_1 and a, if the line $\frac{x - x_1}{a} = y + 2 = \frac{z + 1}{-1}$ and the plane $3x - 4y + 2z = 12$

i. intersect at a unique point **ii.** do not intersect.

2 marks

End of question and answer book for the
2025 Kilbaha VCE Specialist Mathematics Trial Examination 1

Kilbaha Education (Est. 1978) (ABN 47 065 111 373) **PO Box 3229** **Cotham Vic 3101** **Australia**	**PayID: 47065111373** **Email: kilbaha@gmail.com** **Tel: +61 413 425 374** **Web: https://kilbaha.com.au**

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Specialist Mathematics
Examination 1
2025 Formula Sheet

You may keep this Formula Sheet.

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Mensuration

area of a circle segment	$\frac{r^2}{2}\left(\theta-\sin\left(\theta\right)\right)$	volume of a sphere	$\frac{4}{3}\pi r^3$
volume of a cylinder	$\pi r^2 h$	area of a triangle	$\frac{1}{2}bc\sin\left(A\right)$
volume of a cone	$\frac{1}{3}\pi r^2 h$	sine rule	$\frac{a}{\sin\left(A\right)}=\frac{b}{\sin\left(B\right)}=\frac{c}{\sin\left(C\right)}$
volume of a pyramid	$\frac{1}{3}Ah$	cosine rule	$c^2=a^2+b^2-2ab\cos\left(C\right)$

Algebra, number and structure (complex numbers)

$z=x+yi=r\left(\cos\left(\theta\right)+i\sin\left(\theta\right)\right)=r\operatorname{cis}\left(\theta\right)$	$\left\|z\right\|=\sqrt{x^2+y^2}=r$	
$-\pi<\operatorname{Arg}\left(z\right)\le\pi$	$z_1z_2=r_1r_2\operatorname{cis}\left(\theta_1+\theta_2\right)$	
$\frac{z_1}{z_2}=\frac{r_1}{r_2}\operatorname{cis}\left(\theta_1-\theta_2\right)$	de Moivre's theorem	$z^n=r^n\operatorname{cis}\left(n\theta\right)$

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Data analysis, probability and statistics

for independent random variables $X_1, X_2 .... X_n$	$\mathrm{E}(aX_1+b)=a\mathrm{E}(X_1)+b$ $\mathrm{E}(a_1X_1+a_2X_2+...+a_nX_n)$ $=a_1\mathrm{E}(X_1)+a_2\mathrm{E}(X_2)+...+a_n\mathrm{E}(X_n)$	
	$\mathrm{Var}(aX_1+b)=a^2\mathrm{Var}(X_1)$ $\mathrm{Var}(a_1X_1+a_2X_2+...a_nX_n)$ $=a_1^2\mathrm{Var}(X_1)+a_2^2\mathrm{Var}(X_2)+...+a_n^2\mathrm{Var}(X_n)$	
for independent identically distributed variables $X_1, X_2 .... X_n$	$\mathrm{E}(X_1+X_2+...+X_n)=n\mu$	
	$\mathrm{Var}(X_1+X_2+...+X_n)=n\sigma^2$	
approximate confidence interval for μ	$\left(\bar{x}-z\frac{s}{\sqrt{n}}, \bar{x}+z\frac{s}{\sqrt{n}}\right)$	
distribution of sample mean $\bar{X}$	mean	$\mathrm{E}(\bar{X})=\mu$
	variance	$\mathrm{Var}(\bar{X})=\frac{\sigma^2}{n}$

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Calculus

$\frac{d}{dx}\left(x^n\right)=nx^{n-1}$	$\int x^n dx=\frac{1}{n+1}x^{n+1}+c\ ,\ n\neq -1$
$\frac{d}{dx}\left(e^{ax}\right)=ae^{ax}$	$\int e^{ax}dx=\frac{1}{a}e^{ax}+c$
$\frac{d}{dx}\left(\log_e(x)\right)=\frac{1}{x}$	$\int \frac{1}{x}dx=\log_e\left(\lvert x\rvert\right)+c$
$\frac{d}{dx}\left(\sin(ax)\right)=a\cos(ax)$	$\int \sin(ax)dx=-\frac{1}{a}\cos(ax)+c$
$\frac{d}{dx}\left(\cos(ax)\right)=-a\sin(ax)$	$\int \cos(ax)dx=\frac{1}{a}\sin(ax)+c$
$\frac{d}{dx}\left(\tan(ax)\right)=a\sec^2(ax)$	$\int \sec^2(ax)dx=\frac{1}{a}\tan(ax)+c$
$\frac{d}{dx}\left(\cot(ax)\right)=-a\,\mathrm{cosec}^2(ax)$	$\int \mathrm{cosec}^2(ax)dx=-\frac{1}{a}\cot(ax)+c$
$\frac{d}{dx}\left(\sec(ax)\right)=a\sec(ax)\tan(ax)$	$\int \sec(ax)\tan(ax)dx=\frac{1}{a}\sec(ax)+c$
$\frac{d}{dx}\left(\mathrm{cosec}(ax)\right)=-a\,\mathrm{cosec}(ax)\cot(ax)$	$\int \mathrm{cosec}(ax)\cot(ax)dx=-\frac{1}{a}\mathrm{cosec}(ax)+c$
$\frac{d}{dx}\left(\sin^{-1}(ax)\right)=\frac{a}{\sqrt{1-(ax)^2}}$	$\int \frac{1}{\sqrt{a^2-x^2}}dx=\sin^{-1}\left(\frac{x}{a}\right)+c,\ a>0$
$\frac{d}{dx}\left(\cos^{-1}(ax)\right)=\frac{-a}{\sqrt{1-(ax)^2}}$	$\int \frac{-1}{\sqrt{a^2-x^2}}dx=\cos^{-1}\left(\frac{x}{a}\right)+c,\ a>0$
$\frac{d}{dx}\left(\tan^{-1}(ax)\right)=\frac{a}{1+(ax)^2}$	$\int \frac{a}{a^2+x^2}dx=\tan^{-1}\left(\frac{x}{a}\right)+c$
	$\int (ax+b)^n dx=\frac{1}{a(n+1)}(ax+b)^{n+1}+c,\ n\neq -1$
	$\int (ax+b)^{-1}dx=\frac{1}{a}\log_e\left(\lvert ax+b\rvert\right)+c$

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Calculus- continued

product rule	$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$
quotient rule	$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$
chain rule	$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$
integration by parts	$\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$
Euler's method	If $\frac{dy}{dx} = f(x),\ x_0 = a$ and $y_0 = b$, then $x_{n+1} = x_n + h$ and $y_{n+1} = y_n + hf(x_n, y_n)$
arc length parametric	$\int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$
surface area Cartesian about the x-axis	$\int_{x_1}^{x_2} 2\pi y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$
surface area Cartesian about the y-axis	$\int_{y_1}^{y_2} 2\pi x\sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$
surface area parametric about the x-axis	$\int_{t_1}^{t_2} 2\pi y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$
surface area parametric about the y-axis	$\int_{t_1}^{t_2} 2\pi x\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Kinematics

acceleration	$a=\frac{d^2x}{dt^2}=\frac{dv}{dt}=v\frac{dv}{dx}=\frac{d}{dx}\left(\frac{1}{2}v^2\right)$	
constant acceleration formulas	$v=u+at$	$s=ut+\frac{1}{2}t^2$
	$v^2=u^2+2as$	$s=\frac{1}{2}(u+v)t$

Vectors in two and three dimensions

$\underset{\sim}{r}(t)=x(t)\underset{\sim}{i}+y(t)\underset{\sim}{j}+z(t)\underset{\sim}{k}$	$\lvert\underset{\sim}{r}(t)\rvert=\sqrt{x(t)^2+y(t)^2+z(t)^2}$
	$\dot{\underset{\sim}{r}}(t)=\frac{d\underset{\sim}{r}}{dt}=\frac{dx}{dt}\underset{\sim}{i}+\frac{dy}{dt}\underset{\sim}{j}+\frac{dz}{dt}\underset{\sim}{k}$
for $\underset{\sim}{r_1}=x_1\underset{\sim}{i}+y_1\underset{\sim}{j}+z_1\underset{\sim}{k}$ and $\underset{\sim}{r_2}=x_2\underset{\sim}{i}+y_2\underset{\sim}{j}+z_2\underset{\sim}{k}$	vector scalar product $\underset{\sim}{r_1}\cdot\underset{\sim}{r_2}=\lvert r_1\rvert\lvert r_2\rvert\cos(\theta)=x_1x_2+y_1y_2+z_1z_2$
	vector cross product $\underset{\sim}{r_1}\times\underset{\sim}{r_2}=\begin{vmatrix}\underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k}\\ x_1 & y_1 & z_1\\ x_2 & y_2 & z_2\end{vmatrix}=(y_1z_2-y_2z_1)\underset{\sim}{i}+(x_2z_1-x_1z_2)\underset{\sim}{j}+(x_1y_2-x_2y_1)\underset{\sim}{k}$
vector equation of a line	$\underset{\sim}{r}(t)=\underset{\sim}{r_1}+t\underset{\sim}{r_2}=(x_1+x_2t)\underset{\sim}{i}+(y_1+y_2t)\underset{\sim}{j}+(z_1+z_2t)\underset{\sim}{k}$
parametric equation of line	$x(t)=x_1+x_2t \quad y(t)=y_1+y_2t \quad z(t)=z_1+z_2t$
vector equation of a plane	$\underset{\sim}{r}(s,t)=\underset{\sim}{r_0}+s\underset{\sim}{r_1}+t\underset{\sim}{r_2}$ $=(x_0+x_1s+x_2t)\underset{\sim}{i}+(y_0+y_1s+y_2t)\underset{\sim}{j}+(z_0+z_1s+z_2t)\underset{\sim}{k}$
parametric equation of a plane	$x(s,t)=x_0+x_1s+x_2t \quad y(s,t)=y_0+y_1s+y_2t \quad z(s,t)=z_0+z_1s+z_2t$
Cartesian equation of a plane	$ax+by+cz=d$

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au

Circular (trigonometric) functions

$\cos^2(x)+\sin^2(x)=1$	
$1+\tan^2(x)=\sec^2(x)$	$\cot^2(x)+1=\operatorname{cosec}^2(x)$
$\sin(x+y)=\sin(x)\cos(y)+\cos(x)\sin(y)$	$\sin(x-y)=\sin(x)\cos(y)-\cos(x)\sin(y)$
$\cos(x+y)=\cos(x)\cos(y)-\sin(x)\sin(y)$	$\cos(x-y)=\cos(x)\cos(y)+\sin(x)\sin(y)$
$\tan(x+y)=\dfrac{\tan(x)+\tan(y)}{1-\tan(x)\tan(y)}$	$\tan(x-y)=\dfrac{\tan(x)-\tan(y)}{1+\tan(x)\tan(y)}$
$\sin(2x)=2\sin(x)\cos(x)$	
$\cos(2x)=\cos^2(x)-\sin^2(x)$ $=2\cos^2(x)-1=1-2\sin^2(x)$	$\tan(2x)=\dfrac{2\tan(x)}{1-\tan^2(x)}$
$\sin^2(ax)=\dfrac{1}{2}(1-\cos(2ax))$	$\cos^2(ax)=\dfrac{1}{2}(1+\cos(2ax))$

End of Formula Sheet

© Kilbaha Education https://kilbaha.com.au
This page must be counted in surveys by Copyright Agency Limited (CAL)
http://copyright.com.au